

ASYMPTOTIC DISTRIBUTIONS OF RANK STATISTICS IN THE CHANGE-POINT PROBLEM

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Summary: Rank based procedures are considered for testing whether a sequence of independent observations are identically distributed against alternatives involving a change in distribution at an unknown point. Asymptotic distribution theory under small parameter changes is derived. A class of rank based control charts is also constructed and shown to contain the procedure of Bhattacharya and Frierson (*Ann. Statist.* 9, pp. 544-554) as a special case.

1. INTRODUCTION

In this paper we derive the asymptotic distribution of rank statistics for testing whether a sequence of observations are identically distributed against alternatives involving a change in distribution at an unknown point. Let the observations be $X_1, \dots, X_N$ and suppose that for some k , $1 \leq k \leq N-1$, $X_1, \dots, X_k$ have a common unknown mean μ ,

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while $X_{k+1}, \dots, X_N$ have a common mean $\mu + \Delta$, where Δ is also unknown. Consider the statistics $V_i = X_i - (X_1 + \dots + X_N)/N$ for $i = 1, \dots, N$. Then $V_1, \dots, V_k$ will have a common mean $\Delta(k-N)/N$ and $V_{k+1}, \dots, V_N$ a common mean $\Delta k/N$. A test of the hypothesis $\Delta = 0$ could therefore be based on some appropriate functional of the sequence $(V_1, \dots, V_N)$. In the particular case of exponential families, MacNeill (1974, where further references may also be found) has demonstrated the appropriateness of the functional $\sum_{i=1}^{N-1} w(N, i) \sum_{k=i+1}^N V_i$ where $w(\cdot, \cdot)$ is some weight function. One of the objects of the present paper is to extend MacNeill's results to some classes of rank statistics. Other non-parametric approaches to the change-point problem may be found in the papers by Pettitt (1979, 1981) and Sen (1977, 1978).

The method used in this paper consists of projecting the quantities V_i into a suitable class of rank statistics and then forming the corresponding functional of the resulting rank statistics. Extensive use is made of the techniques developed by Lombard (1981) in respect of the class of *linear serial rank statistics*. Note that, in view of the natural ordering of the data, the use of serial rank statistics is particularly appropriate in this problem. On the other hand, from a purely methodological point of view, it is shown in section 3 below that the use of serial ranks provides easy access to the corresponding results for the usual linear rank statistics. In this connection we also refer to the recent papers by Mason (1981) and Lombard (1981).

In sections 2 and 3 the asymptotic distributions of the rank statistics are derived and in section 4 these results are applied to tests for a change in distribution. In section 5, we construct a class of control charts and show that it contains the procedure of Bhattacharya and Frierson (1981) as a special case.

We assume familiarity with the concepts of weak convergence of probability measures in the space $D[0,1]$ of functions possessing discontinuities of at most the first kind. The symbol $\Rightarrow$ is used throughout to indicate such convergence. We also assume that the reader is acquainted with the concept of contiguity of sequences of probability measures as developed in the book by Hájek and Šidák (1967). The following notations are also consistently used: For any nonnegative real number x ,

$$(1.1) \quad (x) = \inf\{r: r \geq x, r \in \{0,1,2,\dots\}\}$$

and for any $t \in [0,1]$ and integer $N \geq 1$,

$$(1.2) \quad \tau(N,t) = \sum_{j=1}^{N-1} j I((2j-1)/2N < t \leq (2j+1)/2N) + NI(1-1/2N < t \leq 1)$$

where $I(\cdot)$ is the indicator function. Note, in this connection, that $\lim_{N \rightarrow \infty} \tau(N,t)/N = \lim_{N \rightarrow \infty} \tau(N,t)/N = t$, uniformly in $t \in [0,1]$. Finally, $(W(t), 0 \leq t \leq 1)$ denotes the standard Brownian motion on $[0,1]$ and $(B(t) = W(t) - tW(1), 0 \leq t \leq 1)$ the standard Brownian bridge.

2. WEAK CONVERGENCE THEOREMS FOR LINEAR SERIAL RANK STATISTICS

Let $\Theta = \{\theta\}$ be a subset of the real line containing the point $\theta = 0$ and let $\{f(x;\theta); \theta \in \Theta, x \in (-\infty, \infty)\}$ be a family of probability density functions which satisfies the conditions

A1: $f(x;\theta)$ is absolutely continuous in θ for almost all (Lebesgue) x .

A2: $f(x;0) \equiv \lim_{\theta \rightarrow 0} (f(x;\theta) - f(x;0))/\theta$ exists for almost all (Lebesgue) x .

$$A3: \lim_{\theta \rightarrow 0} I(f; \theta) = I(f; 0) < \infty$$

$$\text{where } I(f; \theta) = \int_{-\infty}^{+\infty} (\dot{f}(x; \theta) / f(x; \theta))^2 f(x; \theta) dx$$

$$\text{and } \dot{f}(x; \theta) \equiv \frac{\partial}{\partial \theta} f(x; \theta).$$

In particular, these are satisfied in the following cases:

(i) Translation families: $f(x; \theta) = f(x - \theta)$ with f absolutely continuous,

$$\int (f'(x) / f(x))^2 f(x) dx < \infty \quad \text{and} \quad \Theta = (-\infty, +\infty).$$

(ii) Scale families: $f(x; \theta) = e^\theta f(xe^\theta)$ with f absolutely continuous,

$$\int (xf'(x) / f(x))^2 f(x) dx < \infty \quad \text{and} \quad \Theta = (-\infty, +\infty).$$

(iii) Mixture families: $f(x; \theta) = \theta g(x) + (1 - \theta)h(x)$ with

$$\int (g^2(x) / h(x)) dx < \infty \quad \text{and} \quad \Theta = [0, 1].$$

(iv) Lehmann families: $\int_{-\infty}^x f(t; \theta) dt = (F(x))^{1+\theta}$ with F an

arbitrary distribution function and $\Theta = (-\infty, +\infty)$.

Also let $X_1, \dots, X_N$ be independent random variables such that X_i has density function $f(x; \theta(N, i))$, $i=1, \dots, N$ and $\{\theta(N, i); 1 \leq i \leq N; N \geq 1\}$ is an array of elements of Θ satisfying the conditions

$$B1: \{\max \theta^2(N, i); 1 \leq i \leq N\} \rightarrow 0$$

$$B2: \sum_{i=1}^N \theta^2(N, i) \rightarrow b^2 / I(f; 0), \quad 0 \leq b^2 < \infty$$

$$B3: \sum_{i=1}^{\tau(N, t)} \theta(N, i) / N^{\frac{1}{2}} \rightarrow h(t); \quad (h(t), 0 \leq t \leq 1) \in D[0, 1].$$

If, in particular, the hypothesis...

$$(2.1) \quad H_0: \theta(N,1) = \dots = \theta(N,N) = 0$$

is true for every $N \geq 1$, then conditions B above hold with $b = 0$ and $h(t) \equiv 0$. In everything that follows, the symbols P , E and var denote respectively probability, expectation and variance under H_0 . Under H_0 in (2.1), the random variables $X_1, \dots, X_N$ are identically distributed and if $F(x) = \int_{-\infty}^x f(t;0)dt$, the random variables $U_i = F(X_i)$, $i=1, \dots, N$ are independently uniformly distributed over $(0,1)$. The order statistics of $U_1, \dots, U_N$ will be denoted by $U_{N:1}, \dots, U_{N:N}$.

Finally, let $\{a(i,j), 1 \leq j \leq i; i \geq 1\}$ be an array of constants (scores) and $\varphi: (0,1) \rightarrow (-\infty, \infty)$ a function such that

$$C1: \int_0^1 \varphi(u) du = 0; \quad 0 < A^2 \equiv \int_0^1 \varphi^2(u) du < \infty$$

$$C2: \sum_{j=1}^i a(i,j) = 0 \quad \text{for } i \geq 2$$

$$C3: \int_0^1 (a(i, (u_i)) - \varphi(u))^2 du \rightarrow 0$$

If, in particular, $a(i,j) \equiv a(i,j,\varphi)$ where

$$(2.2) \quad a(i,j,\varphi) = E(\varphi(U_{i,j})), \quad j=1, \dots, i$$

then C3 holds.

The serial rank of X_i is given by

$$(2.3) \quad R_i = \sum_{j=1}^i I(X_j \leq X_i), \quad 1 \leq i \leq N$$

and it is well-known that $R_1, \dots, R_N$ are, under H_0 , independently distributed with

$$(2.4) \quad P(R_i = j) = 1/i \quad \text{for } j=1, \dots, i.$$

Condition C3 therefore implies that

$$(2.5) \quad \text{var}(a(i, R_i)) \rightarrow A^2$$

and

$$(2.6) \quad \text{var}(a(i, R_i) - a(i, R_i, \varphi)) \rightarrow 0.$$

Consider now, for $r=1, \dots, N-1$ and $N \geq 2$, the statistics

$$(2.7) \quad T(N, r) = \frac{r}{N} \sum_{i=1}^N \varphi(U_i) - \sum_{i=1}^r \varphi(U_i); \quad T(N, 0) = T(N, N) = 0.$$

$$(2.8) \quad S(N, r, \varphi) = r \sum_{i=r+1}^N a(i, R_i, \varphi)/(i-1); \quad S(N, 0, \varphi) = S(N, N, \varphi) \equiv 0.$$

$$(2.9) \quad S(N, r) = r \sum_{i=r+1}^N a(i, R_i)/(i-1); \quad S(N, 0) = S(N, N) \equiv 0.$$

Lemma 2.1 For $r=1, \dots, N$, under H_0 in (2.1) and conditions C1, C2:

$$E(T(N, r) - S(N, r, \varphi))^2 = E(T(N, r))^2 - E(S(N, r, \varphi))^2.$$

Proof: Let $c(N, i) = (\frac{r}{N} - 1)I(1 \leq i \leq r) + \frac{r}{N}I(r+1 \leq i \leq N)$. Then

$$T(N, r) = \sum_{i=1}^N c(N, i) \varphi(U_i) \quad \text{and, applying Lemma 4.2 of Lombard}$$

(1981), the projection of this into the class of linear serial rank statistics based on $X_1, \dots, X_N$ is exactly $S(N, r, \varphi)$.

Hence the Lemma.

Lemma 2.2. For $r=1, \dots, N$, let $F(N, r)$ be the σ -field generated by $U_{r+1}, \dots, U_{r+r}$, $U_{r+1}, \dots, U_N$. Then under H_0 in (2.1) and condition C2, the sequences $(T(N, r)/r, F(N, r); 1 \leq r \leq N)$ and $(S(N, r)/r, F(N, r); 1 \leq r \leq N)$ are reversed martingales.

Proof. The result for the T -sequence is well-known. To prove it for the S -sequence note that

$$E(S(N, r)/r | F(N, r+1)) = E(a(r+1, R_{r+1})/r | U_{r+1:1}, \dots, U_{r+1:r+1}) \\ + \sum_{i=r+2}^N E(a(i, R_i)/(i-1) | U_{r+1:1}, \dots, U_{r+1:r+1}, U_{r+2}, \dots, U_N).$$

The first term on the right-hand side is zero by condition C2 and the fact that, under H_0 , R_{r+1} is independent of $U_{r+1:1}, \dots, U_{r+1:r+1}$. Since for $i \geq r+2$,

$$R_i = \sum_{j=1}^{r+1} I(U_j \leq U_i) + \sum_{j=r+2}^i I(U_j \leq U_i) \\ = \sum_{j=1}^i I(U_{r+1:j} \leq U_i) + \sum_{j=r+2}^i I(U_j \leq U_i)$$

is $F(N, r+1)$ -measurable, the second term on the right-hand side above is just

$$\sum_{i=r+2}^N a(i, R_i)/(i-1) = S(N, r+1)/(r+1).$$

Lemma 2.3 Under H_0 and conditions C1, C2:

- (i) $\text{var}(T(N, r)/r) - \text{var}(T(N, r+1)/(r+1)) \leq A^2/r^2$
- (ii) $\sum_{r=1}^N r \text{var}(T(N, r)/r) \leq N A^2/2$
- (iii) $\text{var}(S(N, r, \varphi)/r) - \text{var}(S(N, r+1, \varphi)/(r+1)) = \text{var}(a(i, R_i, \varphi))/2$

$$(iv) \quad \sum_{r=1}^N r \operatorname{var}(S(N, r, \varphi)/r) = \sum_{i=2}^N (i/(i-1)) \operatorname{var}(a(i, R_i, \varphi))/2.$$

$$(v) \quad \operatorname{var}(S(N, r, \varphi)/r - S(N, r)/r) = \operatorname{var}(S(N, r+1, \varphi)/(r+1) - S(N, r+1)/(r+1)) \\ = \operatorname{var}(a(r+1, R_{r+1}, \varphi) - a(r+1, R_{r+1}))/r^2.$$

$$(vi) \quad \sum_{r=1}^N r \operatorname{var}(S(N, r, \varphi)/r - S(N, r)/r) = \sum_{i=2}^N (i/(i-1)) \operatorname{var}(a(i, R_i, \varphi) - a(i, R_i))/2.$$

(vii) $\operatorname{var}(T(N, r) - S(N, r))/N \rightarrow 0$ for any sequence $\{r = r(N)\}$ such that $r/N \rightarrow t$, $0 < t < 1$.

Proof: (i)-(vi). Straightforward computation, using (2.4) and the independence of $R_1, \dots, R_N$.

(vii) Using Lemma 2.1, (2.5) and (2.6)

$$\begin{aligned} & \operatorname{var}(T(N, r) - S(N, r)) \\ & \leq 2(\operatorname{var}(T(N, r)) - \operatorname{var}(S(N, r, \varphi))) + 2 \operatorname{var}(S(N, r, \varphi) - S(N, r)) \\ & = 2(A^2(1-r/N)r - r^2 \sum_{i=r+1}^N \operatorname{var}(a(i, R_i, \varphi))/(i-1)^2) \\ & \quad + 2r^2 \sum_{i=r+1}^N \operatorname{var}(a(i, R_i, \varphi) - a(i, R_i))/(i-1)^2 \\ & = o(N), \text{ since} \\ & r^2 \sum_{i=r+1}^N 1/(i-1)^2 \sim Nt(1-t). \end{aligned}$$

Lemma 2.4 Let conditions A and B hold and let

$$L_N = \prod_{i=1}^N (f(X_i; \theta(N, i))/f(X_i; 0)).$$

Then

$$(2.10) \quad \log L_N - \sum_{i=1}^N \theta(N, i)(\dot{f}(X_i; 0)/f(X_i; 0)) + b^2/2 \rightarrow 0 \text{ in}$$

probability and the sequence of distributions of $(X_1, \dots, X_N)$ is contiguous to its sequence of distributions under H_0 .

Proof: Analogous to that of Theorem VI 2.1 of Hájek & Šidák (1967 pp. 211-213, cf. also pp. 238-239).

Theorem 2.1 Let conditions A, B and C hold. Then

(2.11) $\max\{|T(N,r) - S(N,r)|; 1 \leq r \leq N-1\}/AN^{\frac{1}{2}} \rightarrow 0$ in probability and

(2.12) $(S(N, \tau(N,t))/AN^{\frac{1}{2}}, 0 \leq t \leq 1) \Rightarrow (B(t) + e(\varphi, f)(h(t) - th(1)), 0 \leq t \leq 1)$ where $\tau(N,t)$ is given by (1.2),

(2.13) $e(\varphi, f) = \int_0^1 \varphi(u)(\dot{f}(F^{-1}(u); 0)/f(F^{-1}(u); 0))du/A$ and F^{-1} is the inverse of F .

Proof: Let $\varepsilon > 0$ be given and set $x_r = T(N,r)/r - S(N,r,\varphi)/r, y_r = S(N,r,\varphi)/r - S(N,r)/r$. From Lemma 2.2, Chow's (1960) martingale inequality and the inequality $(x_r + y_r)^2 \leq 2x_r^2 + 2y_r^2$ we obtain under H_0 that

$$\begin{aligned} & P\{\max\{|T(N,r) - S(N,r)|; 1 \leq r \leq N-1\} > \varepsilon AN^{\frac{1}{2}}\} \\ &= P\{\max\{|x_r + y_r|; 1 \leq r \leq N-1\} > \varepsilon AN^{\frac{1}{2}}\} \\ &\leq 2 \sum_{r=1}^{N-1} \{r^2((Ex_r^2 - Ex_{r+1}^2) + (Ey_r^2 - Ey_{r+1}^2))\} / \varepsilon^2 A^2 N. \end{aligned}$$

Applying Lemmas 2.1 and 2.3 as well as the relations (2.5) and (2.6) to the last expression above, we see that (2.11) holds under H_0 . In virtue of the contiguity established in Lemma 2.4, it follows that (2.11) holds in general.

In order to establish (2.12) it suffices, in view of (2.11) to show that

$$(T(N, \tau(N,t))/AN^{\frac{1}{2}}, 0 \leq t \leq 1) \Rightarrow (B(t) + e(\varphi, f)(h(t) - th(1)), 0 \leq t \leq 1).$$

But it is straightforward to show, using Le Cam's third lemma (Hájek & Šidák (1967 p. 208)) under the stated conditions, that

$$(Z(N,t) = \frac{\tau(N,t)}{\sum_{i=1}^N \varphi(U_i)} / AN^{\frac{1}{2}}, 0 \leq t \leq 1) \Rightarrow (W(t) + e(\varphi, f)h(t), 0 \leq t \leq 1).$$

Hence,

$$\begin{aligned} (T(N, \tau(N,t)) / AN^{\frac{1}{2}}, 0 \leq t \leq 1) &= (Z(N,t) - (\tau(N,t)/N)Z(N,1), 0 \leq t \leq 1) \\ &\Rightarrow (W(t) + e(\varphi, f)h(t) - t(W(1) + e(\varphi, f)h(1)), 0 \leq t \leq 1) \\ &= (B(t) + e(\varphi, f)(h(t) - th(1)), 0 \leq t \leq 1) \quad \blacksquare \end{aligned}$$

Next, let $\{c(i), i \geq 1\}$ be a sequence of constants and put, for $j = 1, 2, \dots$

$$(2.14) \quad \bar{c}(j) = \frac{1}{j} \sum_{i=1}^j c(i); \quad B^2(j) = \sum_{i=1}^j (c(i) - \bar{c}(j))^2.$$

Assume also that the following conditions hold:

$$D1: \max\{(c(i) - \bar{c}(N))^2; 1 \leq i \leq N\} / B^2(N) \rightarrow 0$$

$$D2: \sum_{i=1}^{N(t)} (c(i) - \bar{c}(N(t))) \theta(N, i) / B(N) \rightarrow \mu(t); (\mu(t), 0 \leq t \leq 1) \in D[0, 1].$$

D3: $B^2(N(t)) / B^2(N) \rightarrow \Psi(t)$ for some strictly increasing continuous function $\Psi: [0, 1] \rightarrow [0, 1]$ with $\Psi(0) = 0$ and $\Psi(1) = 1$, and some non-decreasing left-continuous function $N: [0, 1] \rightarrow \{0, 1, \dots, N\}$ with $N(1) = 1$. Then, for the statistics

$$(2.15) \quad S(M) = \sum_{i=2}^M (c(i) - \bar{c}(i-1)) a(i, R_1), \quad M \geq 2; \quad S(0) = S(1) \equiv 0,$$

we have the following result.

Theorem 2.2: Let conditions A, B1, B2, C, and D hold. Then $(S(N(t))/AB(N), 0 \leq t \leq 1) \Rightarrow (W(\Psi(t)) + e(\varphi, f)\mu(t), 0 \leq t \leq 1)$.

Proof: A version of the theorem was proved by Lombard (1981, Lemma 4.3 and Theorem 2) in the special case where $f(x; \theta) = f(x - \theta)$ and $\theta(N, i) = \beta d_i / (\sum_{j=1}^N d_j^2)^{1/2}$ with β a constant and $\{d_i, i \geq 1\}$ a sequence of constants satisfying the condition $\max\{d_i^2; 1 \leq i \leq N\} / (\sum_{i=1}^N d_i^2)^{1/2} \rightarrow 0$. In view of Lemma 2.4 above the same proof, with only some minor modifications, holds under the more general conditions given above. We omit the details. ■

3. WEAK CONVERGENCE THEOREMS FOR LINEAR RANK STATISTICS

Let R_{Ni} be the rank of X_i among $X_1, \dots, X_N$

$$(3.1) \quad R_{Ni} = \sum_{j=1}^N I(X_j \leq X_i); \quad i = 1, \dots, N, \quad N \geq 2$$

and let $R_{Nk:1}, \dots, R_{Nk:k}$ denote the order statistics of $R_{N1}, \dots, R_{Nk}$ for $k = 2, \dots, N$. Introduce, in addition to (2.7) - (2.9), the statistics

$$(3.2) \quad L(N, r, \varphi) = \sum_{i=r+1}^N a(N, R_{Ni}, \varphi); \quad L(N, N, \varphi) \equiv 0$$

and

$$(3.3) \quad L(N, r) = \sum_{i=r+1}^N a(N, R_{Ni}); \quad L(N, N) \equiv 0.$$

Note that under conditions C1 and C2,

$$L(N, 0, \varphi) = \sum_{i=1}^N a(N, i, \varphi) = N \int_0^1 \varphi(u) du = 0$$

and

$$L(N, 0) = \sum_{i=1}^N a(N, i) = 0.$$

Lemma 3.1 For $r=1, \dots, N$, under H_0 in (2.1) and conditions $C1, C2$:

$$E(L(N, r, \varphi) - S(N, r, \varphi))^2 = E(L(N, r, \varphi))^2 - E(S(N, r, \varphi))^2.$$

Proof: Immediate from the fact (Lombard (1981), p. 138)) that $S(N, r, \varphi)$ is the projection of $L(N, r, \varphi)$ into the class of linear serial rank statistics.

Lemma 3.2 For $r=1, \dots, N$, let $G(N, r)$ be the σ -field generated by $R_{Nr+1}, \dots, R_{Nr+r}, R_{Nr+1}, \dots, R_{NN}$. Then, under H_0 in (2.1) and condition $C2$ the sequences $(S(N, r)/r, G(N, r); 1 \leq r \leq N)$ and $(L(N, r)/r, G(N, r); 1 \leq r \leq N)$ are reversed martingales.

Proof: For the S -sequence the result follows from Lemma 2.2 upon noting that $S(N, r)$ is $G(N, r)$ -measurable and that $G(N, r) \subset F(N, r)$ for $r=1, \dots, N$. For the L -sequence, we have

$$\begin{aligned} & E\left(\sum_{i=r+1}^N a(N, R_{Ni})/r \middle| R_{Nr+1}, \dots, R_{Nr+r}, R_{Nr+1}, \dots, R_{NN}\right) \\ &= \sum_{i=r+2}^N a(N, R_{Ni})/r + E(a(N, R_{Nr+1})/r \middle| R_{Nr+1}, \dots, R_{Nr+r+1}) \\ &= \sum_{i=r+2}^N a(N, R_{Ni})/r + \sum_{i=1}^r a(N, R_{Ni})/r(r+1) \\ &= \sum_{i=r+2}^N a(N, R_{Ni})/(r+1). \end{aligned}$$

Lemma 3.3

- (i) $\text{var}(L(N, r, \varphi)/r) - \text{var}(L(N, r+1, \varphi)/(r+1)) \leq (N/(N-1))\text{var}(a(N, R_N, \varphi))/r^2$
- (ii) $\sum_{r=1}^{N-1} r \text{var}(L(N, r, \varphi)/r) = (N^2/2(N-1))\text{var}(a(N, R_N, \varphi))$

- (iii) $\text{var}(L(N, r, \varphi)/r - L(N, r)/r) - \text{var}(L(N, r+1, \varphi)/(r+1) - L(N, r+1)/(r+1))$
 $\leq (N/(N-1))\text{var}(a(N, R_N, \varphi) - a(N, R_N))/r^2$
- (iv) $\sum_{r=1}^{N-1} r \text{var}(L(N, r, \varphi)/r - L(N, r)/r) = (N^2/2(N-1))\text{var}(a(N, R_N, \varphi) - a(N, R_N)).$

Proof: Straightforward computation using Theorem II.3.1c of Hájek & Šidák (1967) together with (2.3) and (2.4) above. ■

Theorem 3.1 Let conditions A, B and C hold. Then

$$(3.4) \quad \max\{|L(N, r) - S(N, r)|; 1 \leq r \leq N-1\}/AN^{1/2} \rightarrow 0 \text{ in}$$

probability and

$$(3.5) \quad (L(N, \tau(N, t))/AN^{1/2}, 0 \leq t \leq 1) \Rightarrow (B(t) + e(\varphi, f)(h(t) - th(1)), 0 \leq t \leq 1)$$

with $e(\varphi, f)$ given by (2.13).

Proof: Set $x_r = S(N, r)/r - S(N, r, \varphi)/r$, $y_r = S(N, r, \varphi)/r - L(N, r, \varphi)/r$, $z_r = L(N, r, \varphi)/r - L(N, r)/r$. Then $S(N, r) - L(N, r) = x_r + y_r + z_r$ and $(x_r + y_r + z_r)^2 \leq 4x_r^2 + 4y_r^2 + 4z_r^2$. Using lemmas 2.1-2.3 and 3.1-3.3, the proof is entirely analogous to that of theorem 2.1. We omit the details. ■

Remarks

1. In the case of the Wilcoxon statistic, which is obtained by setting $a(N, i) = \frac{i}{N+1} - \frac{N}{2}$ in (3.3), the result of Theorem 3.1 provides rigorous justification for the approximations used by Pettitt (1979).
2. Let $G^*(N, r)$ be the σ -field generated by $R_{N1}, \dots, R_{Nr}$. Then $(L(N, r)/(N-r), G^*(N, r); r=1, \dots, N)$ is a martingale.

see Sen (1978) p. 232.

That this result is equivalent to the one derived in Lemma 3.2 above may be seen by noting the invariance of the distribution of $(R_{N1}, \dots, R_{NN})$ under the transformation $Y_i = X_{N-i+1}$, $i=1, \dots, N$. Sen (1978 p. 232) proves (3.5) above by invoking "some standard arguments", presumably the martingale invariance principle of Brown (1971). This involves the verification of conditions on the distribution of $(L(N, r)/(N-r); r=1, \dots, N)$ which guarantee tightness. On the other hand, the proof of (3.5) given above relies on the projection result in Lemma 3.1 to demonstrate the asymptotic equivalence of $(L(N, \tau(N, t)), 0 \leq t \leq 1)$ and a process with a known limit distribution.

4. RANK TESTS FOR A CHANGE IN DISTRIBUTION

Let $X_1, \dots, X_N$ be independent random variables with density function $f(x_1; \theta(N, 1)), \dots, f(x_N; \theta(N, N))$ and consider, in addition to H_0 in (2.1), the hypotheses

$$(4.1) \quad H_k: \theta(N, 1) = \dots = \theta(N, k) = 0; \theta(N, k+1) = \dots = \theta(N, N) = \beta/N^{\frac{1}{2}} (\beta \neq 0, k=1, \dots, N-1)$$

and

$$H_*: \bigcup_{k=1}^{N-1} H_k.$$

A rank test of H_0 against any given H_k may be based on either of the statistics $S(N, k)$ or $L(N, k)$ defined in (2.9) and (3.3) respectively. In order to test H_0 against H_* we could, following MacNeill (1974), consider as test statistics the weighted sums

$$(4.2) \quad Q_*(N) = \sum_{k=1}^{N-1} w(N, k) (S(N, k))^2$$

or

$$(4.3) \quad V_*(N) = \sum_{k=1}^{N-1} w(N,k) (L(N,k))^2$$

where $\{w(N,k); 1 \leq k \leq N, N \geq 2\}$ is an array of constants.

Theorem 4.1 Let $q(t)$, $0 \leq t \leq 1$ be a nonnegative function satisfying the condition

$$(4.4) \quad \int_0^1 t(1-t)q(t)dt < \infty$$

and set, in (4.2) and (4.3),

$$(4.5) \quad w(N,k) = \frac{(2k+1)/2N}{(2k-1)/2N} \int_{(2k-1)/2N}^{(2k+1)/2N} q(t)dt; \quad k=1, \dots, N-1; \quad N \geq 2.$$

Let conditions A and C of section 2 hold. Then, under $H_\tau(N, \xi)$ with $0 \leq \xi < 1$,

$$(4.6) \quad Q_*(N)/A^2N \Rightarrow \int_0^1 q(t)(B(t) + H(t, \xi))^2 dt$$

where

$$(4.7) \quad H(t, \xi) = \beta e(\phi, f)((t-\xi)I(t \geq \xi) - t(1-\xi))$$

and $e(\phi, f)$ is given by (2.13).

The same result holds with $V_*(N)$ replacing $Q_*(N)$.

Proof: The proof rests on an application of Theorem 4.2 of Billingsley (1968). For every $0 < \varepsilon < \frac{1}{2}$, set $A(\varepsilon) = (0, \varepsilon) \cup (1-\varepsilon, 1)$. Since

$$(H(t, \xi))^2 \leq (t(1-t))^2 \leq t(1-t) \quad \text{for } 0 < \xi \leq 1; \quad 0 \leq t \leq 1,$$

it follows from (4.4) that

$$E\left\{\int_{A(\varepsilon)} q(t)(B(t) + H(t, \xi))^2 dt\right\} \leq 4 \int_{A(\varepsilon)} q(t) \cdot t(1-t) dt \rightarrow 0 \text{ as } \varepsilon \rightarrow 0.$$

Consequently, for every $0 \leq \xi < 1$

$$(4.8) \quad \int_{\varepsilon}^{1-\varepsilon} q(t)(B(t) + H(t, \xi))^2 dt \rightarrow \int_0^1 q(t)(B(t) + H(t, \xi))$$

in probability as $\varepsilon \rightarrow 0$.

Recalling the definition of $S(N, r)$ from (2.9), set

$$S_N(t) = S(N, r(N, t)) / AN^{\frac{1}{2}}, \quad 0 \leq t \leq 1$$

and notice that

$$Q_*(N) / A^2 N = \int_0^1 q(t)(S_N(t))^2 \cdot dt.$$

Now, under $H_{\tau}(N, \xi)$, it is easy to verify conditions B of section 2 with

$$h(t) = \beta(t - \xi)I(t \geq \xi),$$

so that by (2.12)

$$S_N(\cdot) \Rightarrow B(\cdot) + H(\cdot, \xi).$$

Since, for every $0 < \varepsilon < \frac{1}{2}$,

$$\int_{\varepsilon}^{1-\varepsilon} q(t) dt < \infty,$$

it follows that

$$(4.9) \quad \int_{\varepsilon}^{1-\varepsilon} q(t)(S_N(t))^2 dt \Rightarrow \int_{\varepsilon}^{1-\varepsilon} q(t)(B(t) + H(t, \xi))^2 dt$$

for every $0 < \varepsilon < \frac{1}{2}$.

Finally, under H_0 , using (2.5) and the inequality

$$\sum_{i=k+1}^N (i-1)^{-2} \leq \int_{k+1}^{N+1} (x-1)^{-2} \cdot dx ,$$

it is easy to show that for some positive constant K , not dependent upon t ,

$$E(S_N(t))^2 \leq K \frac{\tau(N,t)}{N} (1 - \frac{\tau(N,t)}{N}) \rightarrow Kt(1-t) ,$$

uniformly in $t \in [0,1]$ as $N \rightarrow \infty$.

Thus, by Chebyshev's inequality, under H_0 ,

$$(4.10) \quad \lim_{\epsilon \rightarrow 0} \limsup_{N \rightarrow \infty} P \left\{ \int_{A(\epsilon)} q(t)(S_N(t))^2 dt > \eta \right\} = 0 \quad \text{for}$$

every $\eta > 0$.

In view of contiguity, however, this same result is also true under $H_{\tau(N,\xi)}$ for $0 < \xi < 1$. Now (4.6) follows from (4.8), (4.9) and (4.10) upon application of the earlier quoted theorem of Billingsley (1968).

Since the proof of (4.6) with $V_*(N)$ in place of $Q_*(N)$ is entirely similar, we omit the details. ■

The distribution of $\int_0^1 q(t)(B(t))^2 \cdot dt$ has been partially tabled for $q(t) = 1$, $q(t) = (t(1-t))^{-1}$ and $q(t) = (k+1)t^k$ for a range of values of k . See MacNeill (1974) for details and further references.

For testing H_0 against H_* with β restricted to be positive say, we could use the statistics

$$(4.11) \quad Q(N) = \sum_{k=1}^{N-1} w(N,k) S(N,k)$$

or

$$(4.12) \quad V(N) = \sum_{k=1}^{N-1} w(N,k) L(N,k)$$

Theorem 4.2 Let $q(t)$, $0 \leq t \leq 1$ be a nonnegative function satisfying the condition

$$(4.13) \quad \int_0^1 (t(1-t))^{\frac{1}{2}} q(t) dt < \infty$$

and let $w(N,k)$ be given by (4.5). Let conditions A and C of section 2 hold. Then, under $H_{\tau(N,\xi)}$ with $0 < \xi \leq 1$,

$$(4.14) \quad Q(N)/AN^{\frac{1}{2}} \rightarrow \int_0^1 q(t)(B(t) + H(t,\xi)) dt$$

with $H(t,\xi)$ given by (4.7).

The same result is true with $V(N)$ in place of $Q(N)$. We omit the proof, which is similar to that of theorem 4.1.

Remarks

1. If, instead of (4.13), the function $q(\cdot)$ is assumed to be U-shaped and square-integrable, then Sen (1978 Theorem 6.1) has shown that under H_0

$$(q(t)L(N,\tau(N,t))/AN^{\frac{1}{2}}, 0 \leq t \leq 1) \Rightarrow (q(t)B(t), 0 \leq t \leq 1).$$

Since the function

$$x \in D[0,1] \rightarrow \int_0^1 x(t) dt$$

is continuous in the Skorokhod topology, it follows for $V(N)$ in (4.12) that

$$V(N) = \int_0^1 q(t)L(N,\tau(N,t))dt/AN^{\frac{1}{2}} \Rightarrow \int_0^1 q(t)B(t)dt.$$

This is precisely the result in Theorem 4.2 which pertains to $V(N)$. Note, however, that (4.13) is a much weaker condition than square-integrability so that for an interesting U-shaped weight function such as $q(t) = (t(1-t))^{-\frac{1}{2}}$ we cannot have recourse to Sen's result, but must rely on Theorem 4.2.

2. Suppose the X_i have distributions which lie in the exponential family, i.e.

$$f(x; \theta_1) = g(x) \exp(\theta_1 \varphi(x) - \psi(\theta_1)).$$

The (Bayesian) likelihood ratio tests derived by MacNeill are based on statistics of the form

$$\int_0^1 q(t) (T_N(t))^2 dt,$$

with T_N given by (2.7). It follows from (2.11) and Theorem 4.1 that the rank tests based on $Q_*(N)$ and $V_*(N)$ have the same asymptotic power as these likelihood ratio tests.

3. Under H_0 in (2.1) the X_i are i.i.d. Under a more general null hypothesis, as for instance in the regression change-point problem, the X_i would not be i.i.d. We do not consider this problem in the present paper. In this regard, we refer the reader to Sen (1980), for some nonparametric tests based on supremum functionals.

5. NONPARAMETRIC CONTROL CHARTS

Next, consider the construction of a nonparametric control chart for detecting a change in the level of a process. In order to be realistic we consider runs of finite length, N , say. Thus $X_1, \dots, X_N$ are observed sequentially and the chart should signal to stop as soon as possible after the level of the process has

changed. Here we consider a *repeated significance test* procedure based on statistics

$$(5.1) \quad Q(N, M) = \sum_{k=1}^{M-1} w(N, k) S(M, k), \quad M=2, \dots, N; \quad Q(N, 0) = Q(N, 1) \equiv 0,$$

which may be recognized as sequential versions of those in (4.11). As the data become available we calculate $Q(N, 2), Q(N, 3), \dots$ and stop at the first $M \geq 2$ for which $Q(N, M) \geq d(N, M)$, where $d(N, 1), \dots, d(N, N)$ are constants. In order to investigate the asymptotic properties of such a stopping rule we need to know the asymptotic joint distribution of $Q(N, 2), \dots, Q(N, N)$. Consider, in particular, the weights

$$(5.2) \quad w(N, k) = (k/N)^s, \quad s \geq -1.$$

Substituting into (5.1) and interchanging summations, (recall the definition of $S(N, k)$ from (2.9)). We obtain

$$(5.3) \quad Q(N, M) = N^{-s} \sum_{i=1}^M \left(\sum_{k=1}^{i-1} k^{s+1} / (i-1) \right) a(i, R_i).$$

Theorem 5.1. Let conditions A and C of section 2 hold and let $H(N\xi)$ in (4.1) be true for some fixed $0 \leq \xi < 1$. Then, for $Q(N, M)$ given by (5.3),

$$(5.4) \quad ((s+2)(2s+3))^{\frac{1}{2}} Q(N, (Nt)) / AN^{\frac{3}{2}}, \quad 0 \leq t \leq 1 \Rightarrow (W(t^{2s+3}) + \beta e(\varphi, f) G(t, \xi)), \quad 0 \leq t \leq 1$$

where, for $s > -1$

$$(5.5) \quad G(t, \xi) = \frac{(2s+3)^{\frac{1}{2}}}{(s+1)} \xi^{s+2} ((t/\xi)^{s+1} - 1) I(\xi \leq t \leq 1)$$

and for $s = -1$

$$(5.6) \quad G(t, \xi) = \xi \log(t/\xi) I(\xi \leq t \leq 1)$$

with $\xi \log(1/\xi)$ defined by continuity at $\xi = 0$. The constant $e(\varphi, f)$ is given in (2.13).

Proof: Let $c(1) = 0$ and $c(i) = \sum_{k=1}^{i-1} k^s$ for $i \geq 2$. Some algebraic manipulation shows that $c(i) - \bar{c}(i-1) = \sum_{k=1}^{i-1} k^{s+1}/(i-1)$.

Thus the statistics $N^s Q(N, M)$, which do not depend on N , are of the form given in (2.15) and hence Theorem 2.2 may be applied. Using the asymptotic relation

$$\sum_{k=1}^j k^s \sim \int_1^j x^s dx \quad \text{for } j \rightarrow \infty \quad (s \geq -1 \text{ fixed})$$

we readily obtain (see (2.14))

$$c(i) \sim (i^{s+1}/(s+1)) I(s > -1) + (\log i) I(s = -1),$$

$$\bar{c}(N) \sim (N^{s+1}/(s+1)(s+2)) I(s > -1) + (\log N) I(s = -1),$$

$$B^2(N) \sim (N^{2s+3}/(s+2)^2(2s+3)) I(s \geq -1),$$

and condition D1 holds. Some further calculation shows that condition D2 also holds with $\mu(t) = \beta G(t, \xi)$ and $G(t, \xi)$ given by (5.5) and (5.6). Finally we note that $B^2((Nt))/B^2(N) \rightarrow t^{2s+3}$, so that condition D3 also holds with $\psi(t) = t^{2s+3}$ and $N(t) = (Nt)$. ■

Remarks

1. Consider, in particular, the case $s = -1$ in the above Theorem. Then (5.4) becomes

$$\left(\sum_{i=2}^{(Nt)} a(i, R_i) / AN^{\frac{1}{2}}, 0 \leq t \leq 1 \right) \Rightarrow (W(t) + \beta e(\varphi, f) G(t, \xi), 0 \leq t \leq 1)$$

with $G(t, \xi)$ given by (5.6). This generalizes Theorem 1 of Bhattacharya and Frierson (1981) who considered the special case $a(i, j) = j/i + (i+1)/2i$. Their control chart is therefore but one of many which may be generated using different weight functions in the repeated significance test procedure described above.

2. Within the Bayesian framework developed by MacNeill (1974, pp. 950-952), the weight function which yields the Battacharya-Frierson control chart corresponds to an improper prior density

$$P(\xi \in dt) = \frac{dt}{t}, \quad 0 < t \leq 1$$

for the change-point ξ . Using $s=0$ in (5.2) leads to the "flat" prior

$$P(\xi \in dt) = dt, \quad 0 \leq t \leq 1$$

which is supposed to reflect prior ignorance about ξ . In this case we obtain from theorem 4.3 that

$$\left(\sqrt{3} \sum_{i=2}^{(Nt)} ia(i, R_i) / AN^{\frac{3}{2}}, \quad 0 \leq t \leq 1 \right) \Rightarrow (W(t^3) + \beta e(\varphi, f) G(t, \xi), \quad 0 \leq t \leq 1)$$

where from (5.5) with $s=0$,

$$G(t, \xi) = \sqrt{3} \xi^2 (t/\xi - 1) I(\xi \leq t \leq 1).$$

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